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INTRODUCTION



Thinking about climate change: look up and look around!

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ABSTRACT

We introduce this special issue on *Thinking about Climate Change* by reflecting on the role of psychology in responding adaptively to catastrophic global threats. By way of illustration we compare the threat posed by climate change with the extinction-level threat considered in the recent film *Don't Look Up* [McKay, A. (Director). (2021). *Don't Look Up* [Film]. Hyperobject Industries]. Human psychology is a critical element in both scenarios. The papers in this special issue discuss the importance of clear communication of scientific information, the dangers of misinformation and the possible role played by motivated reasoning, all themes that are taken up in the film. Ultimately, though, it is not enough to consider psychological factors in isolation: we must also acknowledge that cognitive flaws and psychological motivations are exploited by vested interests that profit from delaying climate action. A global response to a global crisis requires us to 'look up' to recognise the threat and to 'look around' to go beyond specialist disciplines and national boundaries.

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'If we can't all agree at the bare minimum that a giant comet the size of Mount Everest is not a fucking good thing, then what the hell happened to us?'

– Randall Mindy

Is humanity capable of responding adaptively to catastrophic global threats such as climate change or (as in the film *Don't Look Up*; McKay, 2021) approaching comets? To what extent is our response to such threats compromised by our thoughts and beliefs? Adam McKay, the director of *Don't Look Up*, has described it as 'a disaster movie in which people don't necessarily believe that the disaster is coming'. The quotation above is from a scientist in the film who is outraged by people's failure to believe in the existence of the comet or to treat the threat it poses with the appropriate

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level of seriousness. Many of the themes of the film are familiar to climate communicators and scientists working on the psychology of climate change. Is the problem in both cases a failing of human psychology? Is there a fundamental problem in the way we think about climate change? Each of the articles in this special issue relates to this question, in one way or another, probing our difficulty in understanding science, our vulnerability to misinformation and motivated reasoning, and our need for a belief in our own efficacy as a precursor to action.

Students of psychology have long been interested in the extent to which our beliefs, and even our perceptions, are the product of our goals and motives (e.g., Erdelyi, 1974; Festinger, 1957). The title of *Don't Look Up* refers to the slogan chanted by supporters of the (Trump-like) president, who refuse to look up even when the comet is visible to the naked eye. This unwillingness to confront reality could be seen as an extreme form of motivated reasoning, in which the search for and evaluation of evidence is guided by needs, drives, desires, or goals rather than reality (e.g., Kruglanski, 1996; Kunda, 1990). With their chanting and their red baseball caps these followers are asserting their shared identity. Their motivation to maintain a sense of social consensus with an important ingroup may be leading them to adopt beliefs that cohere with that group. A similar motivated reasoning has been proposed to underlie climate denial (e.g., Kahan, 2015). The polarisation and politicisation of science in the film has clear parallels to our response to climate change. Nevertheless, as Molden, Bayes and Druckman (this issue) point out, we need to ask which motivations (if any) are the real drivers of people's information processing. For example, are Republicans more likely than Democrats to question climate science because they seek to reinforce their connection with their social group, or because they believe that the views of people in this group have greater information value than those of people outside those groups? Or do Republicans hold those beliefs because they consume news media that they believe to be more credible in general (but which tend to promote climate scepticism)?

As well as motivations relating to social consensus and information seeking, another possible motivation for downplaying global threats may be fear of the anticipated solutions. Acknowledging anthropogenic climate change implies the need for radical changes to individual behaviour as well as the way societies are structured, which may conflict with one's value system (e.g., Caddick & Feist, this issue; Wolsko et al., 2016). There is also a financial cost of climate change mitigation. Concerns about such costs can be seen at the level of individuals, as in the NOMBA (Not Out of My Bank Account!) phenomenon investigated by Swim et al. (2021). However, such fears are also seen at the corporate and governmental level, with politicians

and business leaders sometimes openly questioning whether it makes economic sense to tackle climate change (e.g., Beament, 2019). Climate mitigation and energy transition will be expensive and there are important social justice issues concerning the division of these costs, but the cost of not tackling the climate crisis is many times greater (Kikstra et al., 2021). In *Don't Look Up*, President Orlean asks at one point, 'What is this going to cost me?', but the film never seriously entertains the notion that the cost of acting would discourage governments from intervening. Perhaps that would have been too absurd – in this respect the film's satire may underplay the true extent of the human capacity to ignore problems until they are no longer solvable.

Of course, several aspects of *Don't Look Up* are exaggerated for comic effect. In addition, various aspects of the comet allegory differ from climate change in important ways: the impact of the comet is a discrete event in the near future that will affect everyone simultaneously. Climate change is more gradual, and its effects are unequally distributed across the planet. Consequently, factors such as temporal and spatial discounting pose additional psychological 'barriers' to climate action (Gifford et al., 2009). The comet scenario is one that should be easier for humans to respond to appropriately than climate change.

Ultimately, however, humanity's response to the comet in the film fails not because of some inherent species-wide flaw in the ability to understand and think about the problem, but because of the greed of entrenched power and commercial interests: a billionaire subverts the attempt to divert the comet so that he can exploit its presumed mineral wealth. While not a perfect analogy, this parallels the way in which powerful companies have sought to delay climate action so as to continue exploiting fossil fuels for profit. In the film, as in life, GDP is prioritised over preservation of life. The techno-solution of seeking to mine the comet rather than diverting it from its path echoes the arguments that suggest that geo-engineering of our climate can be a substitute for reducing emissions.

Indeed, important aspects of the apparent failure of people to think rationally about climate change are not accidents of human psychology, but rather deliberate manipulations of our psychology by vested interests. Bergquist et al. (this issue) report that only 62% of the US population understands that human activities are the primary cause of climate change. This science-public disconnect reflects, in part, the poor communication of scientific topics by mainstream media (also alluded to in the film). However, the problem is not simply poor science communication – it is a deluge of misinformation that causes people to not believe the problem (Lewandowsky, 2021). The prevalence of misinformation continues to be an obstacle to improving public understanding of climate change (Velautham

& Ranney, this issue). Thanks to leaks, freedom of information requests and lawsuits it is now known that fossil fuel companies deliberately spread climate disinformation for decades (e.g., Oreskes & Conway, 2011). This is not to say that cognitive flaws have not been an important contributor to the inadequate response to the threat of climate change. But these psychological aspects have been exploited and weaponised by corporate interests focused on short-term profit and politicians focused on the short-term electoral cycle. In the film, as in life, humanity has been put in a perilous position because those with power have delayed, denied, and misled (Brulle, 2014; Dunlap & McCright, 2011; Lewandowsky, 2021; Supran & Oreskes, 2017).

Approaching comets and the climate crisis are both examples of collective action problems – there is little that individuals can do to meaningfully mitigate against these threats (though this is not to undermine the value of individual action, which may help to facilitate and bring about collective action). A critical challenge for psychology is to investigate how to promote the collective action that will be required to bring about a rapid reduction in emissions and a transition to renewable energy. How do we help people to develop a sense of collective efficacy – a belief in their own power to take part in collective action and a belief in the power of collectives to bring about systemic change? These questions are the subject of ongoing research (Angill-Williams & Davis, this issue; Hornsey, Chapman, & Oelrichs, this issue).

Responding to climate change is a problem that requires contributions from a broad range of disciplines, not only in the physical climate sciences, but also in social sciences and the humanities. Thinking about climate change requires us to think about the human mind not in isolation but as something embedded in social, economic and political structures. We need to ‘look up’ to recognise the impending threat and its real causes, but we also need to look *around* – outside of our specialist disciplines and across national boundaries – to see how we can harness the power of the collective to mount a truly global response to a global crisis.

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